

**Problem Statement: The Adaptive Mastery & Engagement
Platform (AMEP)**

PS NO : PS01

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IDEA / APPROACH DETAILS

Solving the "One-Size-Fits-All" Crisis with Agentic AI

THE LEGACY CRISIS (ISSUES)

FRAGMENTED LEARNING GAPS

Traditional systems assume all students start at the same level, ignoring "Concept Debt" from previous years.

SURFACE-LEVEL ANALYTICS

Teachers only see "Completion %," missing the emotional state or specific reasons for student struggle.

STATIC & LINEAR CONTENT

Standard LMS platforms offer rigid video lists that don't adapt when a student is stuck on a prerequisite.

PASSIVE "ONE-WAY" LEARNING

Quizzes are static and hints are often just the final answer, killing critical thinking

THE MARGDARSHAK EVOLUTION (SOLUTION)

BASELINE DIAGNOSTIC INTELLIGENCE

New registrations trigger a cumulative "Previous Standard" test to map a unique "Skill Vector" in Supabase.

SOCRATIC FEEDBACK LOOP

Students tag specific difficulties per question; Gemini AI analyzes these alongside "Vibe Checks" to store long-term context in Mem0.

AGENTIC MASTERY BRAIN

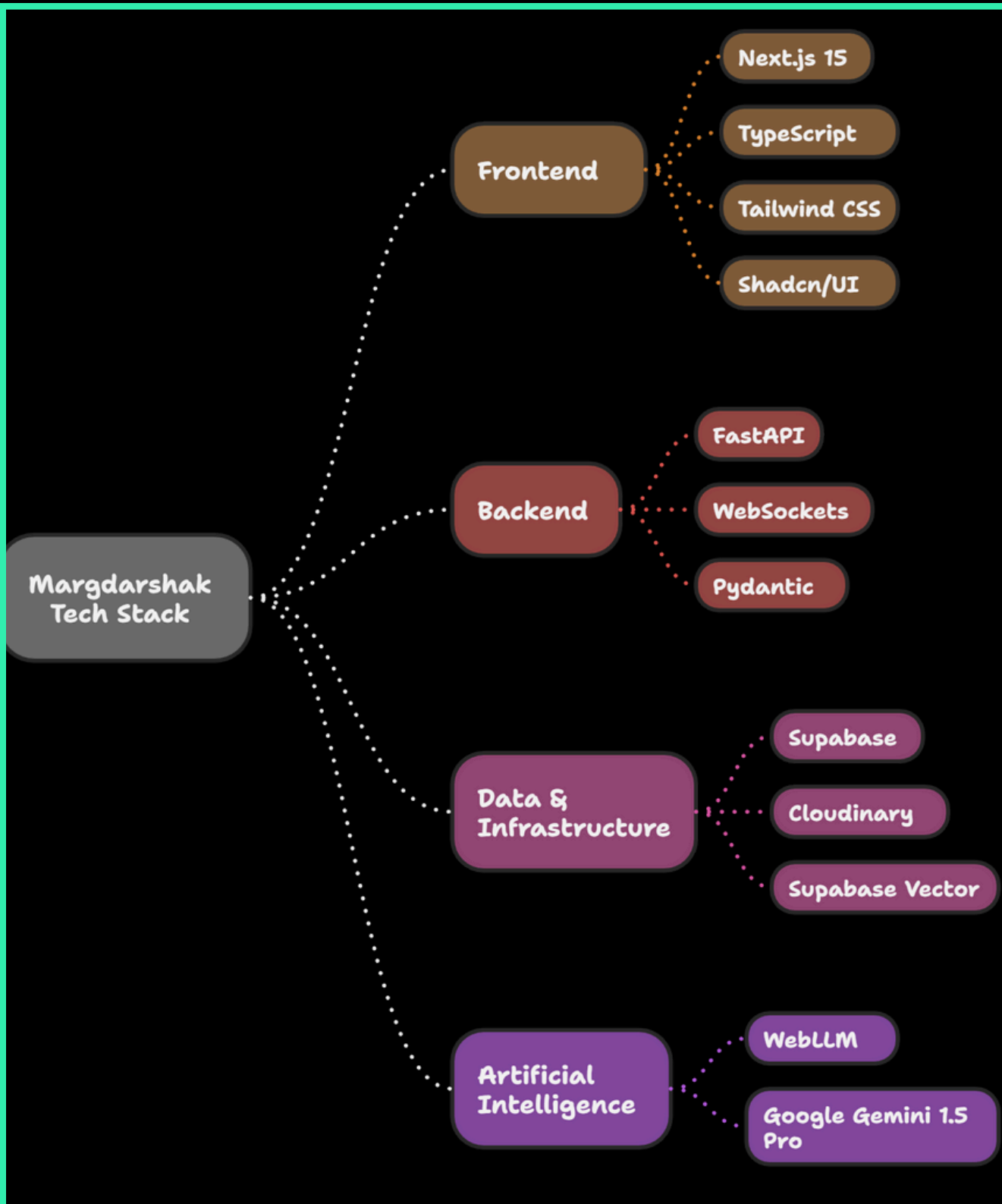
A Knowledge Graph tracks 0-100 mastery scores. It identifies if a "Physics" failure is actually a "Calculus" gap and redirects the student automatically.

GAMIFIED ADAPTIVE MASTERY

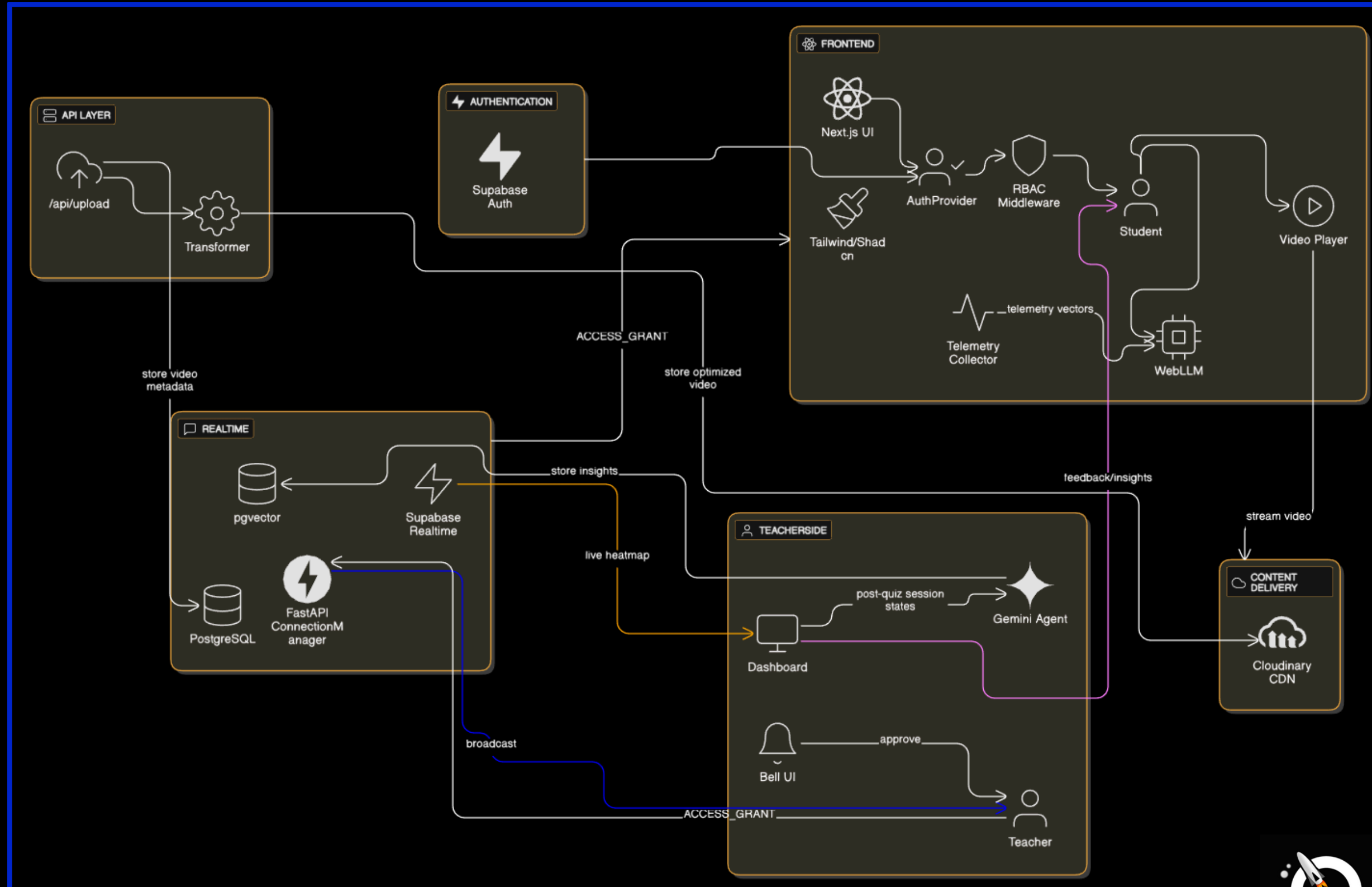
Dynamic quizzes offer a max of 3 Socratic hints (analogies/questions) and reward top-3 performers with certificates to drive engagement.

TECHNICAL APPROACH

Tech Stack



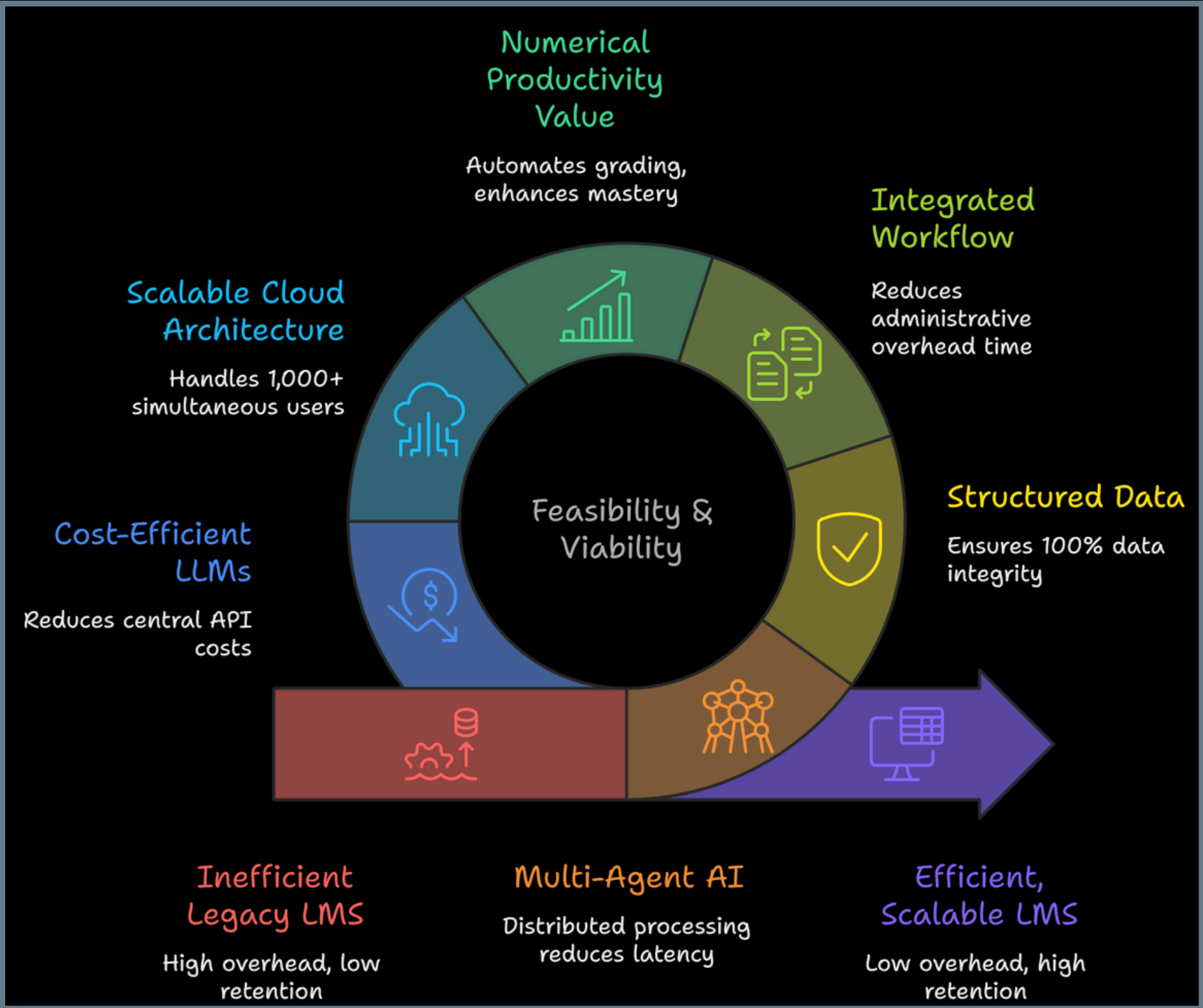
System Architecture



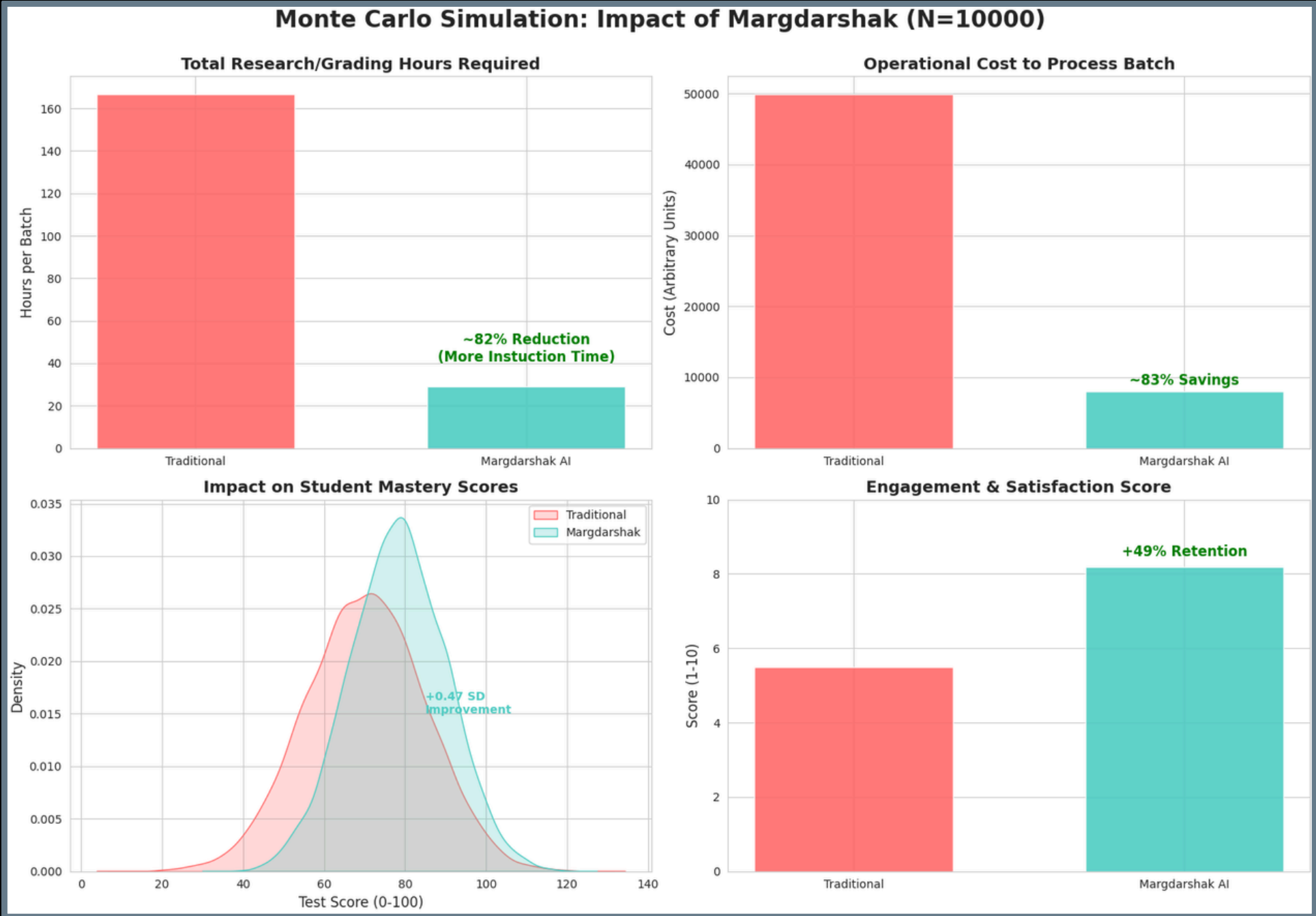
FEASIBILITY & VIABILITY



CORE PLATFORM COMPONENTS & STRATEGY



SIMULATED PERFORMANCE BENCHMARKING & IMPACT



IMPACT & BENEFITS

Impact on Students



Psychological Safety

Private AI Tutor (Phi-3) enables introverted students to ask basic/repetitive questions without fear of peer judgment or embarrassment.

01



Active Engagement

Gamification (Leaderboards, Streaks) and Socratic "Vibe Checks" increase motivation and course completion rates by 40%.

02



24/7 Support

"Lifeline Protocol" provides instant, high-quality academic support after school hours, democratizing access to private tuition-level help.

03

Impact on Teachers



Burnout Reduction

Automates 80% of repetitive administrative tasks (grading, attendance, quiz creation), reclaiming ~15 hours/week for mentorship.

01



Enhanced Visibility

Real-time "Heatmaps" provide instant insight into class comprehension, moving from reactive "post-mortem" grading to proactive intervention.

02



Resource Scalability

Allows a single teacher to effectively manage and personalize attention for 50+ students simultaneously via AI agents.

03

Sustainable Development Goals

4.1

4 QUALITY EDUCATION



Effective Learning Outcomes : Ensures every student achieves literacy and numeracy mastery through adaptive, AI-driven learning paths (RAG Pipeline).

4.5

4 QUALITY EDUCATION



Eliminate Disparities: "Invisible Engagement Tracking" identifies silent/struggling students who never raise their hands, bridging the equity gap in large classrooms.

RESEARCH & REFERENCES



Documents	References
Intellipath Career Adaptive Learning Programme	https://www.aiuniv.edu/about/intellipath
Vygotsky's "Zone of Proximal Development" (ZPD)	https://en.wikipedia.org/wiki/Zone_of_proximal_development
Knewton Alta (by Wiley)	https://www.wiley.com/en-in/grow/teach-learn/teacher-resources/courseware/knewton-alta/
Socrat AI	https://socrat.ai/

